ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT II

Orthographic Projections of Points, Lines & Planes

See a 3-D object as two flat views.

SYLLABUS COVERED

- ▸ Reference planes & quadrants
- ▸ Projections of a point (4 quadrants)
- ▸ Lines ∥ / ⊥ / inclined to reference planes
- ▸ Line inclined to both planes
- ▸ Planes ⊥ both, ∥ one & inclined to other
- ▸ Plane inclined to both reference planes

1. Orthographic Projection – The Basics

DEFINITION

Orthographic projection represents a 3-D object by **two or more 2-D views** obtained by dropping **projectors perpendicular** to the reference planes. The view on the vertical plane is the **Front View (FV)**; on the horizontal plane the **Top View (TV)**.

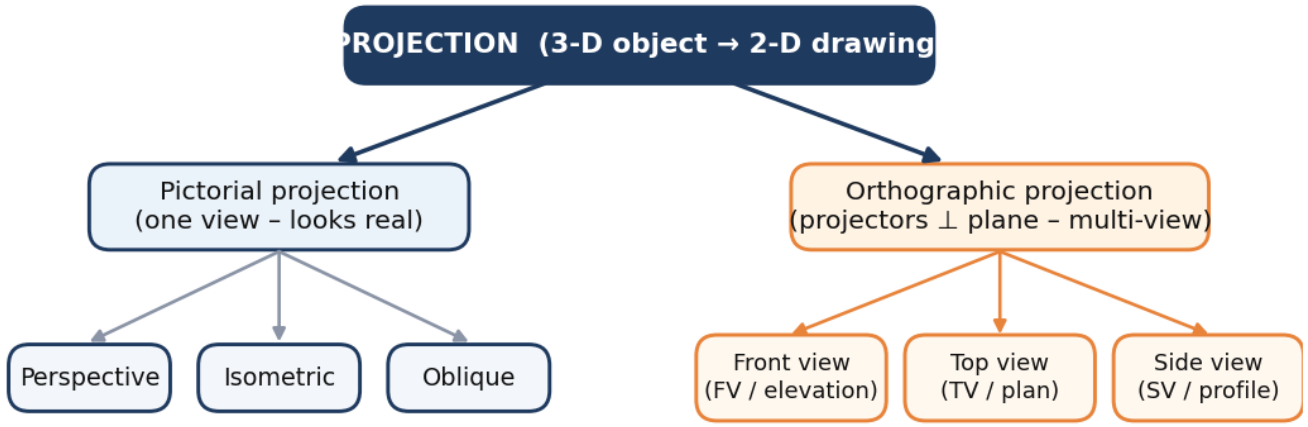


Fig. 1.1 — Classification of projections

Term	Meaning
Plane of projection	Plane on which the view is obtained (HP, VP, PP)
Projector	Imaginary line from the object point ⊥ to the plane
HP / VP / PP	Horizontal Plane / Vertical Plane / Profile (side) Plane
XY line	Line of intersection of HP and VP – the **reference line**
Notation	Object point **A**; FV **a′** (dashed-prime); TV **a**; SV **a″**

EXAM TIP

We follow **First-Angle Projection** (BIS/Indian practice): object lies in the **first quadrant**, between the observer and the plane. In **Third-Angle** projection (USA) the plane lies between observer and object.

2. Reference Planes & Four Quadrants

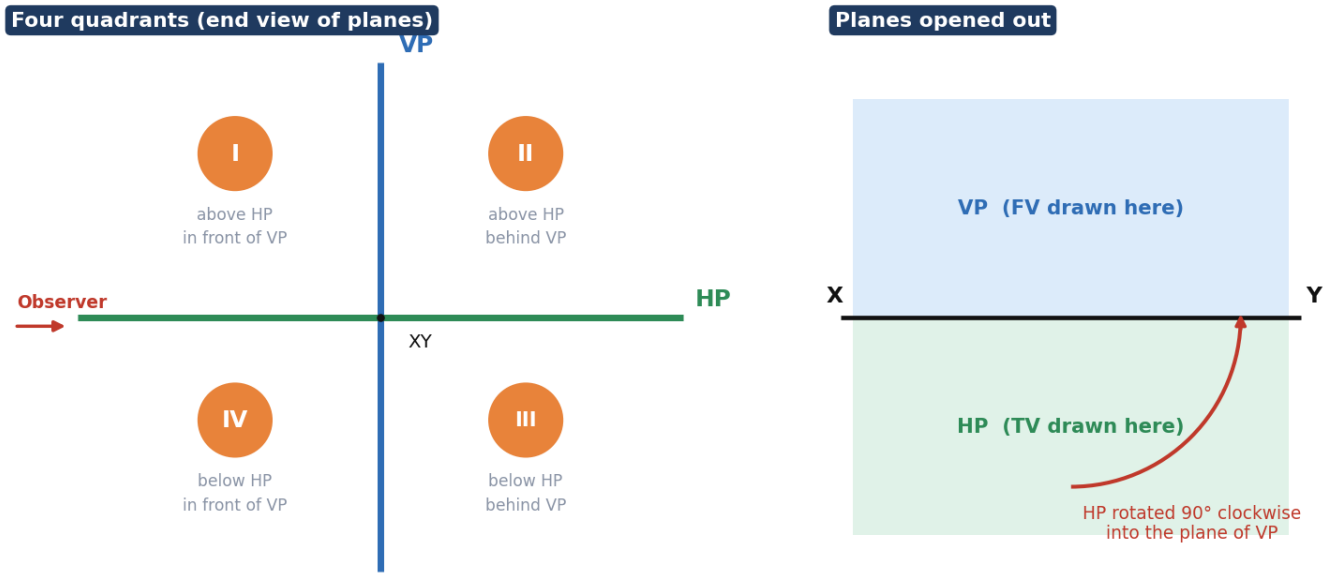


Fig. 2.1 — HP and VP divide space into four quadrants; HP is rotated into the VP to get one flat sheet

- **Importance of XY:** it separates the FV (above XY, on VP) from the TV (below XY, on HP) after rotation; the distance of a view from XY gives the **height** (FV) or **distance in front of VP** (TV).
- The two views of a point always lie on **one vertical projector (⊥ XY)**.
- After rotation: the part of HP **in front** of VP goes **below XY**; the part **behind** VP goes **above XY**.

3. Projection of a Point

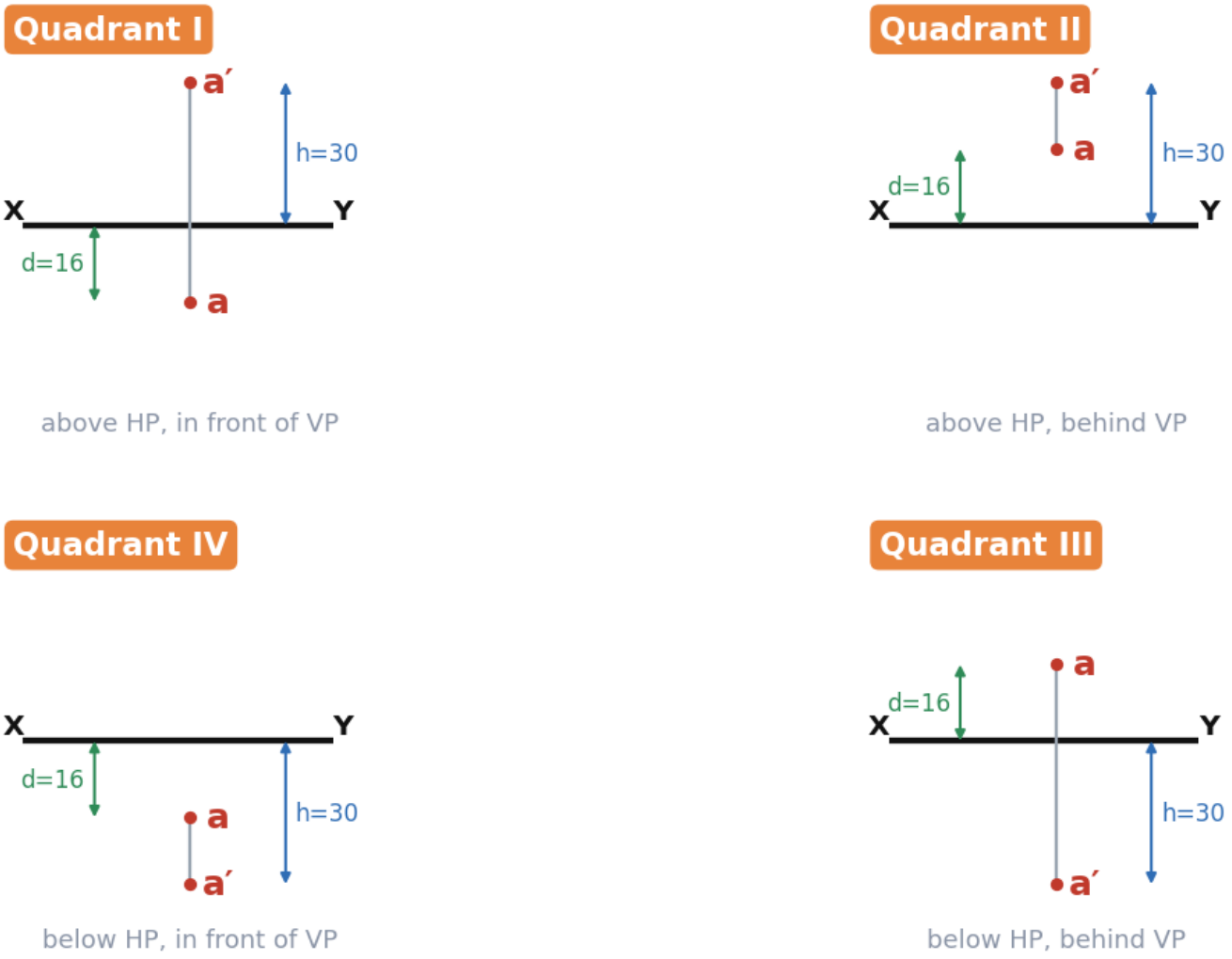


Fig. 3.1 — A point in each quadrant (height above/below HP = 30, distance in front/behind VP = 16)

Quadrant	Position of point	FV a´	TV a
I	above HP, in front of VP	above XY	below XY
II	above HP, behind VP	above XY	**above XY**
III	below HP, behind VP	**below XY**	above XY
IV	below HP, in front of VP	below XY	below XY

KEY POINTS

- FV a´ ⟶ distance from XY = **height of point above/below HP**; TV a ⟶ distance from XY = **distance from VP**.
- Point **in HP**: FV on XY. Point **in VP**: TV on XY. Point in both: both views on XY.
- **Memory trick:** in Quadrants II & IV the two views lie on the *same side* of XY (II – both above, IV – both below).

4. Projections of Straight Lines

A straight line is fixed by the projections of its two end points. Notation: **TL** = true length, **θ** = true inclination to HP, **φ** = true inclination to VP, **α, β** = apparent inclinations in FV and TV.

Position of line	FV	TV
∥ HP and ∥ VP	TL, ∥ XY	TL, ∥ XY
⊥ HP (∥ VP)	TL, ⊥ XY	Point
⊥ VP (∥ HP)	Point	TL, ⊥ XY
Inclined θ to HP, ∥ VP	TL at true θ	Shorter (TL cosθ), ∥ XY
Inclined φ to VP, ∥ HP	Shorter (TL cosφ), ∥ XY	TL at true φ
Inclined to both	Shorter, at α > θ	Shorter, at β > φ

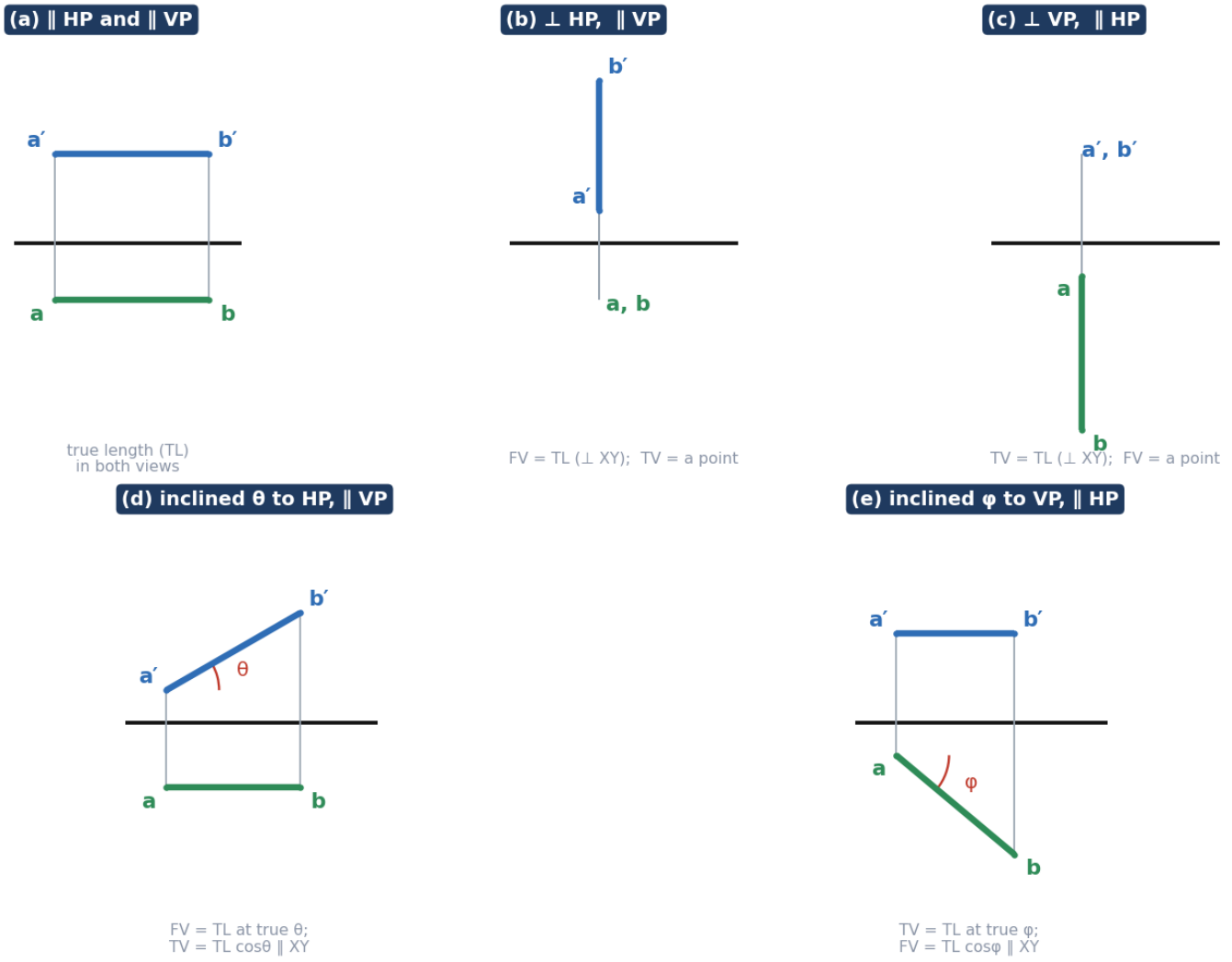


Fig. 4.1 — Lines in special positions

4.1 Line inclined to both reference planes

FORMULAS

TV length = **TL cosθ** | FV length = **TL cosφ** | Height difference of ends = **TL sinθ** | Distance difference = **TL sinφ**

θ + φ ≤ 90° | **α ≥ θ, β ≥ φ** | tanθ = tanα · cosβ | tanφ = tanβ · cosα

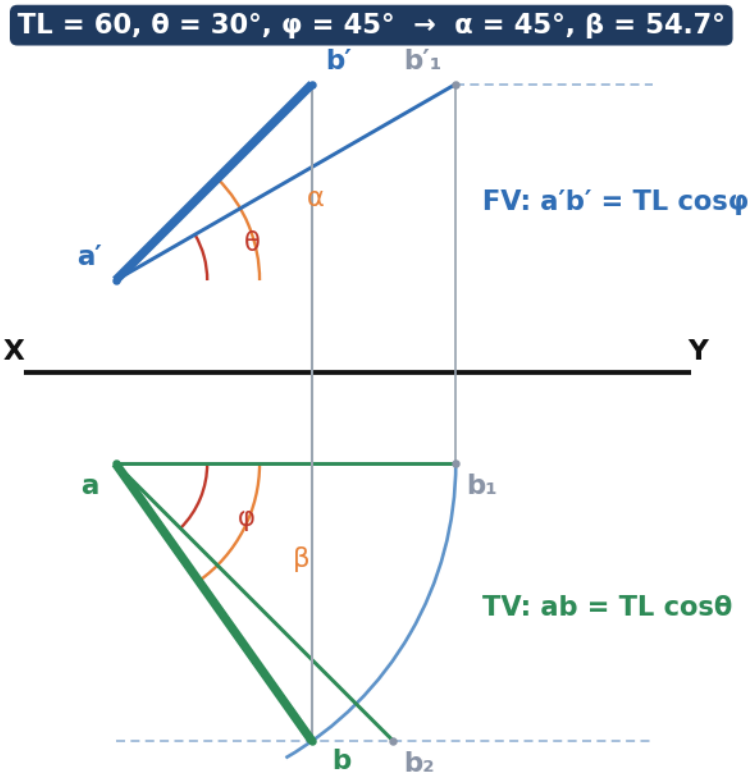


Fig. 4.2 — Line of TL = 60 mm, θ = 30°, φ = 45° (locus method)

PROCEDURE

1. Draw XY. Fix a′ (height) and a (distance). Draw **a′b₁′ = TL at θ**; project TV length **ab₁ = TL cosθ** (horizontal).
2. From a draw **ab₂ = TL at φ** in the TV; the horizontal through b₂ is the **locus of b** (distance = TL sinφ).
3. With centre a and radius **ab₁** draw an arc to cut that locus at **b**.
4. Project b up to the horizontal through b₁′ (locus of b′) to get **b′**. Join a′b′ and ab.

SOLVED EXAMPLE

A line AB, 60 mm long, is inclined 30° to HP and 45° to VP. Find the apparent lengths and angles.

FV = 60 cos45° = **42.4 mm**; TV = 60 cos30° = **52 mm**; height difference = 60 sin30° = **30 mm**; distance difference = 60 sin45° = **42.4 mm**. From the figure: α = **45°**, β = **54.7°**. Check: tanθ = tan45° · cos54.7° = 0.577 ✓.

APPLICATIONS

Line projections are used for **pipelines, roof rafters, struts and trusses** where members lie obliquely in space. **Traces**: the point where a line (extended) meets HP is its **HT** (FV on XY), and meets VP is its **VT** (TV on XY).

5. Projections of Planes

A **plane** is a 2-D figure (triangle, square, pentagon, hexagon, circle…) with negligible thickness. Its position is described by the inclination of its **surface** to the reference planes. Always start with the plane **parallel to a reference plane** (true shape visible) and then tilt it in **stages**.

Position of plane	FV	TV
∥ HP (⊥ VP)	Line ∥ XY	**True shape**
∥ VP (⊥ HP)	**True shape**	Line ∥ XY
⊥ HP and ⊥ VP	Line ⊥ XY	Line ⊥ XY
Inclined θ to HP, ⊥ VP	Line at true θ	Reduced shape
Inclined φ to VP, ⊥ HP	Reduced shape	Line at true φ
Inclined to both	Reduced shape	Reduced shape

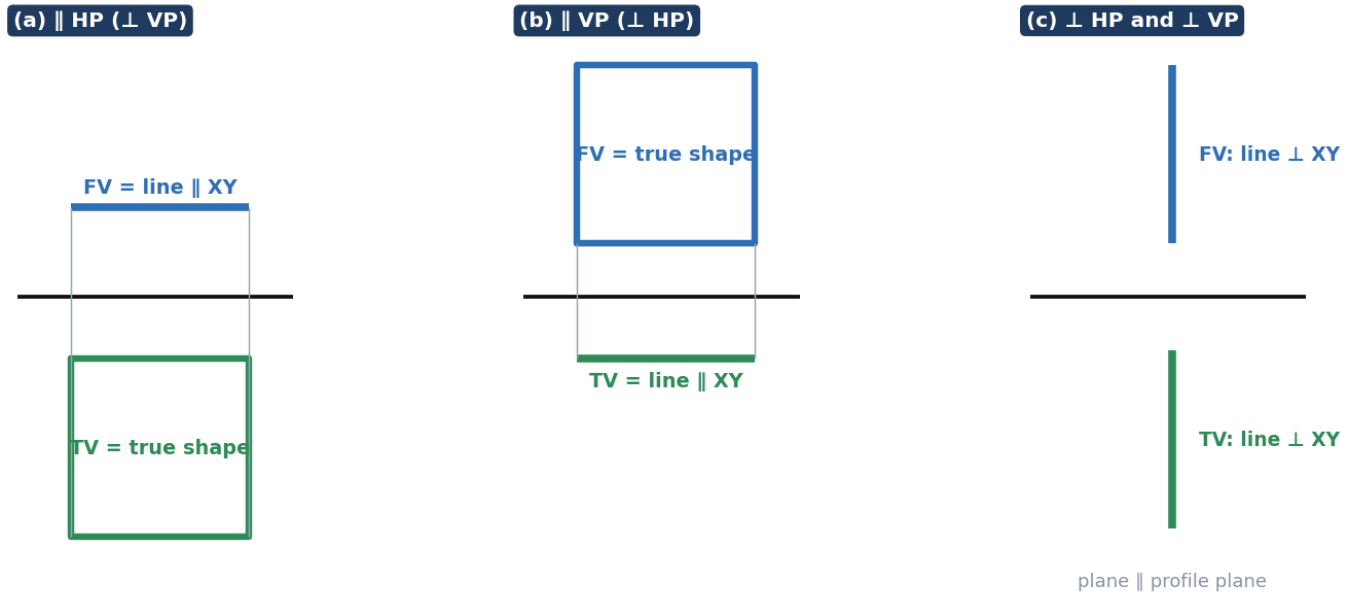


Fig. 5.1 — Planes parallel to a reference plane and perpendicular to both

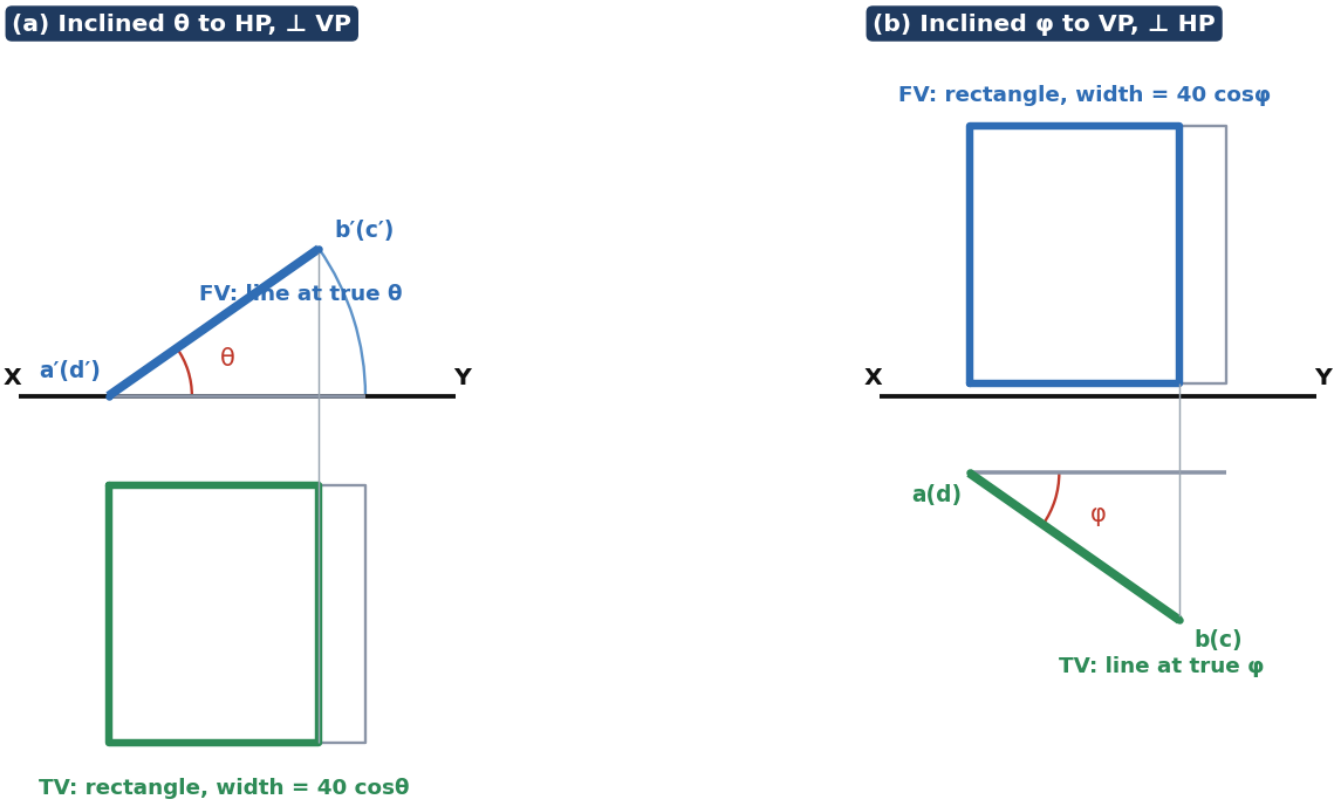


Fig. 5.2 — Plane inclined to one reference plane and perpendicular to the other

FORMULAS

Reduced width = **true width × cos(inclination)** | Area of TV = **true area × cosθ** (plane at θ to HP) | Area of FV = **true area × cosφ**

5.1 Plane inclined to both reference planes

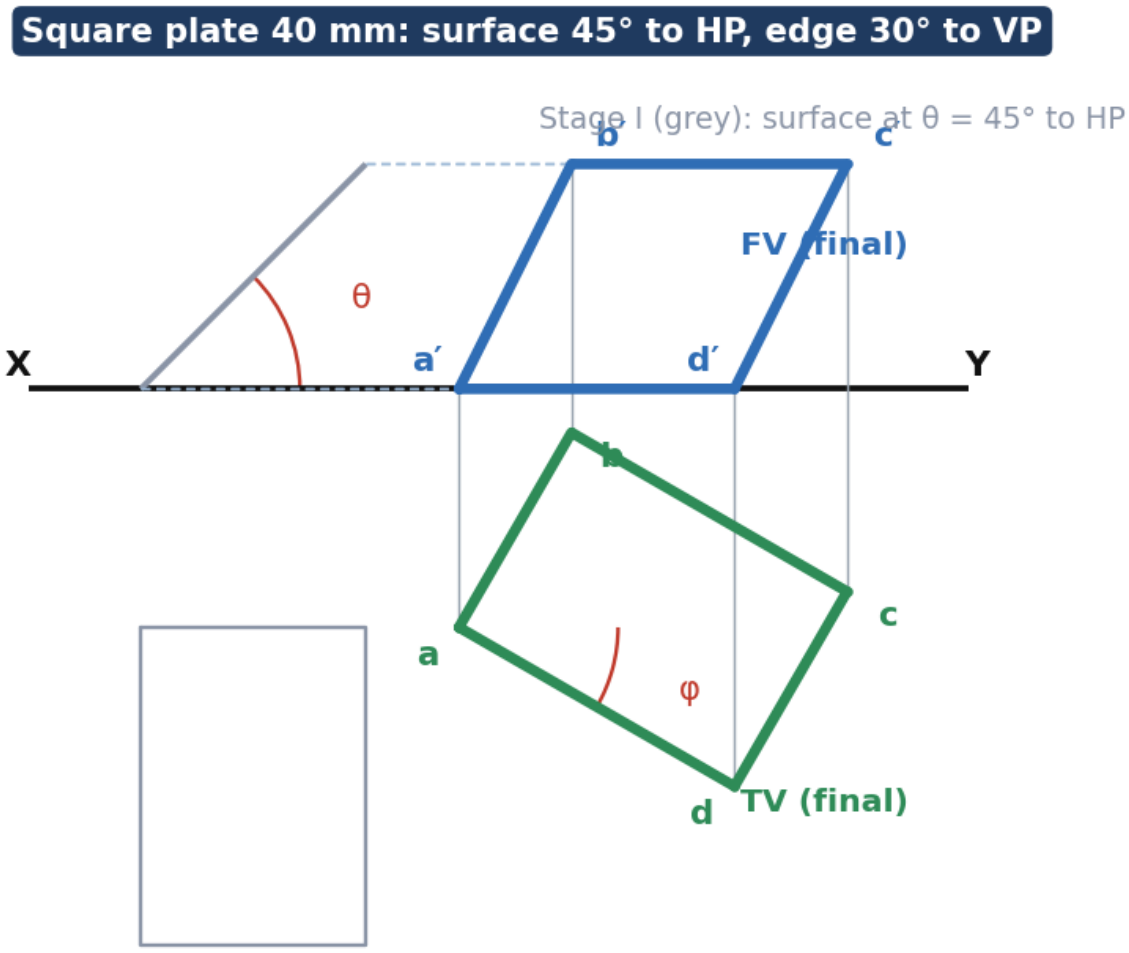


Fig. 5.3 — Two-stage construction for a square plate

PROCEDURE

1. **Stage I:** assume the plane ∥ HP. Draw TV = true shape (one edge ⊥ XY or ∥ XY as given) and FV = line on XY.
2. Tilt the FV about the edge/corner on HP so that it makes **θ with XY**; project the new TV (reduced shape).
3. **Stage II:** redraw the Stage-I TV rotated so that the given edge (or diagonal) makes **φ with XY**. Keep its shape unchanged.
4. Project the final FV: **verticals from the new TV** and **horizontals from the Stage-I FV heights**. Join in order.

EXAM TIP

Golden rule: in a change of position, the **view in which the inclination is shown true is moved rigidly**, and the **other view keeps its heights (or distances)** – they only slide along parallel lines. Hidden edges are drawn as short dashes.

6. Quick Revision & Exam Questions

QUICK SUMMARY

- FV is above XY in Q I & II; TV is below XY in Q I & IV.
- Line ∥ plane → TL in that view; line ⊥ plane → point view.
- TV = TL cosθ, FV = TL cosφ; θ + φ ≤ 90°.
- Plane ∥ plane → true shape; plane ⊥ plane → line view.
- Two-stage method: change **θ first** (FV), then **φ** (TV).

EXAM-ORIENTED QUESTIONS

Q1. Define reference planes and quadrants. Why is the XY line important? **[2]**

Q2. A point P is 30 mm above HP and 25 mm behind VP. Draw its projections and name the quadrant. **[4]**

Q3. Point A is 20 mm below HP and 30 mm in front of VP. Point B is 40 mm above HP and 10 mm behind VP; distance between projectors 50 mm. Draw projections of AB. **[6]**

Q4. A line AB 70 mm long is inclined 30° to HP and 45° to VP. A is 15 mm above HP and 20 mm in front of VP. Draw FV and TV; find α and β. **[8]**

Q5. A 60 mm line is parallel to VP, inclined 40° to HP, end A 10 mm above HP. Draw its projections. **[6]**

Q6. A regular pentagon of side 30 mm has one side on HP; its surface is inclined 45° to HP and ⊥ VP. Draw the projections. **[8]**

Q7. A square plate of 40 mm side has its surface inclined 30° to HP and a diagonal inclined 45° to VP. Draw FV and TV. **[8]**

Q8. Name the position of a plane whose FV and TV are both lines ⊥ XY. **[4]**